

ipv_validation.m

User Guide & Data Derivation Reference

IPV Benchmarking Project · SEE 894 · Simon Fraser University · Spring 2026

Purpose

This script validates indoor photovoltaic (IPV) data digitised from published papers. It applies three internal consistency checks — illuminance (C1), power conversion efficiency (C2), and short-circuit current density (C3) — to determine whether a paper's device can be approved for locus construction. This guide explains every input, output, and derived quantity so that any future user can reproduce or extend the analysis.

1. Overview

The script performs the following steps in sequence:

1. Load the CIE photopic luminosity function $V(\lambda)$ from a standard CSV file.
2. **Load and resample the LED spectral power distribution (SPD)** from a digitised CSV. Apply the correct physical unit conversion based on the `SPD_UNITS` flag.
3. **Load and resample the external quantum efficiency (EQE)** spectrum from a digitised CSV. Convert from % to fraction if needed.
4. **Load, clean, and sign-correct the J–V curve** from a digitised CSV. Extract J_{sc} , V_{oc} , FF, P_{max} , and PCE.
5. **Run Check 1:** Calculate illuminance from the extracted SPD. Must equal 1000 lux $\pm$ 10%.
6. **Run Check 2:** Compare calculated PCE (from J–V and SPD) against the reported PCE. Must agree within $\pm$ 10%.
7. **Run Check 3:** Integrate EQE $\times$ photon flux to obtain J_{sc_int} . Compare against J_{sc} from the J–V curve. Must agree within $\pm$ 10%.
8. **Compute $\lambda_{onset,95\%}$:** the wavelength below which 95% of cumulative LED irradiance is contained.
9. Print a full summary and produce six diagnostic figures.

2. File Requirements

You must supply four CSV files. All four must be placed in the **same folder** as the script.

File / Variable	Description
CIE_sle_photopic.csv	Standard CIE 1931 photopic luminosity function $V(\lambda)$. Column 1 = wavelength (nm), Column 2 = $V(\lambda)$ [0–1]. Provided once with the project — do not change.

spd_file (e.g. s41_spd.csv)	LED spectral power distribution digitised from the paper. Column 1 = wavelength (nm), Column 2 = intensity in the units specified by SPD_UNITS.
ege_file (e.g. s41_ege.csv)	External quantum efficiency spectrum. Column 1 = wavelength (nm), Column 2 = EQE (either 0–1 or 0–100 %; the script auto-detects).
jv_file (e.g. s41_jv.csv)	J–V curve from the paper. Column 1 = voltage (V or mV; auto-detected). Column 2 = current density (uA/cm ² or mA/cm ² ; auto-detected).

Tip: Digitising figures

Use WebPlotDigitizer (<https://automeris.io/WebPlotDigitizer/>). Export each curve as a two-column CSV. Save each file with the naming convention shown above: <paperID>_spd.csv, <paperID>_ege.csv, <paperID>_jv.csv.

3. User-Configurable Parameters

Edit only the block labelled **CHANGE ONLY THIS SECTION** at the top of the script. Do not modify anything else.

Parameter	How to Set It
PAPER_LABEL	A short string identifying the paper (e.g. 's3', 's12'). Used in plot titles and console output only. Any string is valid.
PCE_reported	Power conversion efficiency (%) as stated in the paper. Used in Check 2. Example: 30.3
Voc_reported	Open-circuit voltage (V) as stated in the paper. Used as a fallback only if the digitised J–V curve does not reach J = 0.
E_ref_uWcm2	Total irradiance reported in the paper (μW/cm ²). Used as a reference comparison only — not forced into any calculation.
spd_file	Filename of the SPD CSV (must be in the same folder as the script).
ege_file	Filename of the EQE CSV.
jv_file	Filename of the J–V CSV.
SPD_UNITS	Unit flag for the SPD y-axis. See Section 4 for all valid values.

4. SPD Unit Flags

Choose the flag that matches the y-axis label on the figure you digitised from the paper.

SPD_UNITS value	When to use	Conversion applied
'uW/cm2/nm'	Most common. SPD y-axis is labelled μW cm ⁻² nm ⁻¹ .	Divide by 100 → W/m ² /nm
'mW/m2/nm'	SPD y-axis is labelled mW m ⁻² nm ⁻¹ .	Divide by 1000 → W/m ² /nm
'W/m2/nm'	SPD y-axis is already in SI units.	None (used directly)

'photon_flux'	Axis labelled $\times 10^{12} \text{ cm}^{-2} \text{ s}^{-1} \text{ nm}^{-1}$.	$\times 10^{12} \times 10^4 \times h\nu/\lambda \rightarrow \text{W/m}^2/\text{nm}$
'photon_flux_e13'	Axis labelled $\times 10^{13} \text{ cm}^{-2} \text{ s}^{-1} \text{ nm}^{-1}$.	$\times 10^{13} \times 10^4 \times h\nu/\lambda \rightarrow \text{W/m}^2/\text{nm}$
'photon_flux_e15'	Axis labelled $\times 10^{15} \text{ cm}^{-2} \text{ s}^{-1} \text{ nm}^{-1}$.	$\times 10^{15} \times 10^4 \times h\nu/\lambda \rightarrow \text{W/m}^2/\text{nm}$
'au_rescale'	SPD is in arbitrary units. Script rescales to match E_ref_uWcm2.	Rescale so $\int E \, d\lambda = E_{\text{ref_uWcm2}}/100 \text{ W/m}^2$

Important

The script always converts the SPD to $\text{W/m}^2/\text{nm}$ before any calculation. All downstream quantities (illuminance, irradiance, photon flux, PCE) are derived from this unified internal unit. Never mix units between the flag and the actual CSV values.

5. How Each Check Is Calculated

5.1 Check 1 — Illuminance (C1)

Illuminance is the photometric measure of light intensity perceived by the human eye. It is calculated by weighting the SPD with the CIE photopic luminosity function $V(\lambda)$ and integrating:

$$E_v = 683 \times \int E(\lambda) \cdot V(\lambda) \, d\lambda \quad [\text{lux}]$$

where 683 lm/W is the maximum luminous efficacy (K_m) at 555 nm.

Symbol	Meaning
$E(\lambda) \text{ [W/m}^2/\text{nm]}$	Extracted SPD (after unit conversion). The spectral irradiance of the LED source.
$V(\lambda) \text{ [—]}$	CIE photopic luminosity function, ranging from 0 to 1. Peaks at 1.0 at 555 nm. Interpolated onto the SPD wavelength grid before integration.
683 [lm/W]	Maximum luminous efficacy constant (K_m). Converts radiometric to photometric units.
$E_v \text{ [lux]}$	Resulting illuminance. Must equal 1000 lux $\pm$ 10% for the check to pass.

Why 1000 lux is always the target

The project methodology requires all papers to be compared at 1000 lux. TARGET_LUX is always 1000 regardless of what illuminance the original paper nominally reported. The check verifies that the digitised SPD is physically consistent with this target.

5.2 Check 2 — Power Conversion Efficiency (C2)

PCE is derived entirely from the digitised data — no reported values are forced in:

$$\begin{aligned}
 E_{\text{tot}} &= \int E(\lambda) \, d\lambda \times 100 & [\mu\text{W}/\text{cm}^2] & \quad (\text{integrate SPD, convert } \text{W}/\text{m}^2 \rightarrow \mu\text{W}/\text{cm}^2) \\
 P_{\text{max}} &= \max(V \times J) & [\mu\text{W}/\text{cm}^2] & \quad (\text{maximum power from J-V curve}) \\
 \text{PCE}_{\text{calc}} &= P_{\text{max}} / E_{\text{tot}} \times 100 & [\%] &
 \end{aligned}$$

Symbol	Meaning
$E_{\text{tot}} \, [\mu\text{W}/\text{cm}^2]$	Total irradiance: integral of the extracted SPD over wavelength. Factor $\times 100$ converts W/m^2 to $\mu\text{W}/\text{cm}^2$.
$J \, [\mu\text{A}/\text{cm}^2]$	Current density from the digitised J–V curve, in $\mu\text{A}/\text{cm}^2$.
$V \, [\text{V}]$	Voltage from the digitised J–V curve.
$P_{\text{max}} \, [\mu\text{W}/\text{cm}^2]$	Maximum power density: the peak value of $V \times J$.
$\text{PCE}_{\text{calc}} \, [\%]$	Calculated power conversion efficiency. Compared against $\text{PCE}_{\text{reported}}$.
Deviation	$ \text{PCE}_{\text{calc}} - \text{PCE}_{\text{reported}} / \text{PCE}_{\text{reported}} \times 100 \, \%$. Must be $\leq 10\%$ to pass.

5.3 Check 3 — Integrated Short-Circuit Current Density (C3)

The photocurrent generated by the device is predicted from first principles using EQE and the photon flux of the LED:

$$\begin{aligned}
 \text{Phi}(\lambda) &= E(\lambda) \cdot \lambda / (h \cdot c) & [\text{photons}/\text{m}^2/\text{s}/\text{nm}] \\
 J_{\text{sc_int}} &= q \times \int \text{EQE}(\lambda) \cdot \text{Phi}(\lambda) \, d\lambda \times 100 & [\mu\text{A}/\text{cm}^2]
 \end{aligned}$$

Symbol	Meaning
$E(\lambda) \, [\text{W}/\text{m}^2/\text{nm}]$	Extracted SPD, same as used in C1 and C2.
$\lambda \, [\text{nm}]$	Wavelength in nanometres.
$h \cdot c \, [\text{J} \cdot \text{nm}]$	Planck constant $\times$ speed of light = $1.9864 \times 10^{-25} \, \text{J} \cdot \text{nm}$. Converts energy per photon units.
$\text{Phi}(\lambda) \, [\text{photons}/\text{m}^2/\text{s}/\text{nm}]$	Spectral photon flux of the LED. Derived from the SPD — not measured separately.
$\text{EQE}(\lambda) \, [—]$	External quantum efficiency as a fraction (0–1). Specifies what fraction of incident photons are converted to collected electrons at each wavelength.
$q \, [\text{C}]$	Elementary charge = $1.602 \times 10^{-19} \, \text{C}$. Converts photon count to charge.
$\times 100$	Unit conversion: $\text{A}/\text{m}^2 \rightarrow \mu\text{A}/\text{cm}^2$. ($1 \, \text{A}/\text{m}^2 = 10^{-4} \, \text{A}/\text{cm}^2 = 100 \, \mu\text{A}/\text{cm}^2$)
$J_{\text{sc_int}} \, [\mu\text{A}/\text{cm}^2]$	Predicted short-circuit current from spectral convolution. Integration is over the overlap of the SPD and EQE wavelength ranges only.

Jsc_jv [$\mu\text{A}/\text{cm}^2$]	Measured Jsc from the digitised J–V curve: current density at $V = 0$.
Mismatch	$ Jsc_int - Jsc_jv / Jsc_jv \times 100 \%$. Must be $\leq 10\%$ to pass.

5.4 Spectral Descriptor: $\lambda_{\text{onset},95\%}$

This parameter characterises the long-wavelength reach of the LED spectrum and is the key x-axis variable for locus construction:

$$\int [400 \rightarrow \lambda_{\text{onset},95\%}] E(\lambda) \, d\lambda = 0.95 \times \int [400 \rightarrow \infty] E(\lambda) \, d\lambda$$

In practice: the cumulative integral of $E(\lambda)$ is normalised to $[0, 1]$, then the wavelength at which this cumulative sum equals 0.95 is found by interpolation. This is the wavelength below which 95% of the total irradiance is contained.

The optimal absorber bandgap for a device illuminated by this LED is approximately $hc / \lambda_{\text{onset},95\%}$. This relationship (from Khampa et al. 2026) is why $\lambda_{\text{onset},95\%}$ replaces CCT and CRI as the spectral descriptor for indoor PV benchmarking.

6. J–V Curve Processing Details

The script applies several automatic corrections to the raw digitised J–V data:

Auto-correction	Rule Applied
Voltage unit detection	If $\max(V) > 5$, the data is assumed to be in mV and divided by 1000 to convert to V.
Sorting	Points are sorted by ascending voltage.
Quadrant clipping	Only $V \geq 0$ points are retained (photovoltaic quadrant).
Duplicate removal	Duplicate voltage values are averaged.
Sign convention	If $\text{median}(J)$ at mid-forward-bias voltages is negative, J is flipped. This handles papers that plot current flowing into the device as positive.
Current unit detection	After sign correction, if $\max(J) < 5 \mu\text{A}/\text{cm}^2$, data is assumed to be in mA/cm^2 and multiplied by 1000.
Voc extraction	The first downward zero-crossing of $J(V)$ is found by linear interpolation between the two bracketing points. If J never reaches zero (truncated curve), $Voc_reported$ is used as a fallback.
Jsc extraction	Current density at $V = 0$, obtained by linear interpolation.
FF calculation	$FF = P_max / (Jsc \times Voc)$.

Sign convention note

The script checks the median of J at voltages above 10% of Voc. If this median is negative (current flowing out, reported as negative by some authors), J is negated globally. This is not a blind abs() — it preserves the zero-crossing shape needed for Voc and FF calculations.

7. Output: Console Summary

After running, the console prints a complete validation report. Here is an annotated example for a passing paper:

```
=====
IPV DATA VALIDATION - s3
=====
SPD : s3_spd.csv [380-800 nm, 421 pts]
      Units: uW/cm2/nm -> /100 -> W/m2/nm
EQE : s3_eqe.csv [300-800 nm, 501 pts]
JV  : s3_jv.csv [52 pts] uA/cm2 (used directly)

--- CHECK 1: Illuminance ---
Extracted : 1004.1 lux
Target    : 1000 lux
Deviation  : 0.41% (limit ±10%)
Result     : PASS

--- CHECK 2: PCE ---
Calculated : 42.81%
Reported   : 42.01%
Deviation   : 1.90% (limit ±10%)
Result      : PASS

--- CHECK 3: Integrated Jsc ---
Integration range : 380-740 nm
Jsc (EQE × SPD)   : 122.690 uA/cm2
Jsc (J-V)         : 130.847 uA/cm2
Mismatch          : 6.23% (limit ±10%)
Result            : PASS

--- Spectral Descriptor ---
lambda_onset,95% = 677.2 nm

=====
SUMMARY - s3
=====
C1 Illuminance : 1004.1 lux dev=0.41% PASS
C2 PCE          : calc=42.81% rep=42.01% PASS
C3 Jsc          : mismatch=6.23% PASS
-----
OVERALL : APPROVED for locus construction
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Jsc = 130.847 uA/cm2
Voc = 1.0700 V
FF = 0.8510 (85.1%)
Pmax = 119.322 uW/cm2
E_tot = 278.38 uW/cm2 (extracted)
E_ref = 280.00 uW/cm2 (reported)
lam95 = 677.2 nm
=====
```

8. Output: The Six Diagnostic Figures

Figure	What It Shows
Figure 1 — CIE $V(\lambda)$	The CIE photopic luminosity function from 360–830 nm, with the peak at 555 nm marked. This is a sanity check that the CIE file loaded correctly.
Figure 2 — Extracted SPD	The LED spectral irradiance in $\mu\text{W}/\text{cm}^2/\text{nm}$ after unit conversion and UV clipping. A red dashed line marks $\lambda_{\text{onset},95\%}$. The extracted E_{tot} and the reported E_{ref} are annotated for comparison.
Figure 3 — EQE & Integrated Jsc	Left axis (black): EQE (%) vs wavelength. Right axis (red): cumulative integrated Jsc ($\mu\text{A}/\text{cm}^2$) as wavelength increases. The final value of the red curve equals Jsc_int from Check 3.
Figure 4 — J–V Curve	Left axis (blue circles): current density J ($\mu\text{A}/\text{cm}^2$). Right axis (red dashed): power density $P = V \times J$ ($\mu\text{W}/\text{cm}^2$). The maximum power point is marked with a red square. Jsc, Voc, FF, and PCE_calc are annotated in a text box.
Figure 5 — Illuminance Integrand [C1]	The integrand $683 \times E(\lambda) \times V(\lambda)$ plotted as a filled area. The area under this curve equals the illuminance E_v . Annotated with the computed lux value. Visual diagnostic for Check 1.
Figure 6 — Jsc Integrand [C3]	The integrand $q \times \text{EQE}(\lambda) \times \text{Phi}(\lambda) \times 100$ in $\mu\text{A}/\text{cm}^2/\text{nm}$. The area under this curve equals Jsc_int. Annotated with Jsc_int, Jsc_jv, and their mismatch. Visual diagnostic for Check 3.

9. Derived Data: What Each Number Means

The following table lists every quantity produced by the script, its physical meaning, and its role in the broader IPV benchmarking workflow.

Quantity	Physical Meaning	Role in Analysis
E_v [lux]	Photopic illuminance of the LED source computed from the extracted SPD.	Check 1: must equal 1000 lux $\pm 10\%$. Confirms the digitised SPD represents the correct illumination level.
E_{tot} [$\mu\text{W}/\text{cm}^2$]	Total radiometric irradiance integrated over wavelength.	PCE denominator in Check 2. Also used to confirm consistency with E_{ref} from the paper.
$\lambda_{\text{onset},95\%}$ [nm]	Wavelength below which 95% of total LED irradiance is contained.	X-axis of the PCE locus. Encodes the spectral 'reach' of the LED independently of CCT or CRI.
Jsc_jv [$\mu\text{A}/\text{cm}^2$]	Short-circuit current density from the J–V curve at $V = 0$.	Reference for Check 3. Also recorded as the device's measured Jsc under this LED.
Jsc_int [$\mu\text{A}/\text{cm}^2$]	Short-circuit current density predicted from EQE $\times$ photon flux integration.	Check 3 comparator. Agreement with Jsc_jv confirms that EQE was measured under conditions consistent with the J–V test.

Voc [V]	Open-circuit voltage: voltage at which net current is zero.	Defines the upper voltage limit of power generation. Used to calculate FF.
FF [—]	Fill factor: ratio of maximum power to $J_{sc} \times V_{oc}$.	Measures the 'rectangularity' of the J–V curve. Reflects charge transport quality and recombination losses.
Pmax [$\mu\text{W}/\text{cm}^2$]	Maximum output power density: peak of $V \times J$.	PCE numerator. Directly reflects the useful power the device can deliver under indoor conditions.
PCE_calc [%]	Power conversion efficiency = $P_{max} / E_{tot} \times 100$.	The key performance metric. Used in Check 2 and recorded for the locus if the device passes all checks.

10. Step-by-Step Usage

- Digitise the three curves** from the paper using WebPlotDigitizer: (1) LED SPD, (2) EQE, (3) J–V. Export each as a two-column CSV.
- Place all four CSVs** (CIE_sle_photopic.csv, your SPD, EQE, and J–V files) in the same folder as ipv_validation.m.
- Open ipv_validation.m** in MATLAB and edit only the top configuration block.
- Set PAPER_LABEL** to a short identifier (e.g. 's3').
- Set PCE_reported** and Voc_reported from the paper's reported values.
- Set E_ref_uWcm2** to the irradiance value stated in the paper (used only for reference comparison).
- Set spd_file, eqe_file, jv_file** to your CSV filenames.
- Set SPD_UNITS** by checking the y-axis label of the SPD figure in the paper. See Section 4.
- Run the script** (press F5 or Run in MATLAB). Read the console output. Check all three results are PASS.
- Inspect the six figures.** If any check fails, common causes are: wrong SPD_UNITS, digitising the wrong axis, or an SPD that was not measured at 1000 lux. Refer to Section 11.
- If APPROVED:** record J_{sc_jv} , Voc, FF, PCE_calc, E_{tot} , and $\lambda_{onset,95\%}$ for use in the locus construction script (spectral_projection.m).

11. Troubleshooting

Symptom	Likely Cause and Fix
C1 FAIL — Ev far above 1000 lux	SPD units are wrong. If you set 'uW/cm2/nm' but the axis was actually 'mW/m2/nm', the values are 10× too high. Try changing SPD_UNITS.
C1 FAIL — Ev far below 1000 lux	SPD may be in photon flux units. Try 'photon_flux' or one of the photon_flux_eXX options. Alternatively the digitised curve may be in arbitrary units — use 'au_rescale' with the correct E_ref_uWcm2.

C2 FAIL — PCE_calc << PCE_reported	Check that the J–V voltage is in the correct quadrant ($V > 0$, $J > 0$). The sign-correction should handle most cases, but if the curve only covers $V = 0$ to a small value, Pmax will be underestimated.
C3 FAIL — Jsc_int << Jsc_jv	The EQE was likely measured under different conditions (e.g. AM1.5G monochromatic illumination, not indoor broadband). This is a known limitation; the script correctly flags it. The paper should be excluded.
C3 FAIL — Jsc_int >> Jsc_jv	The EQE or J–V curve may have been digitised from the wrong figure. Verify both CSVs against the paper.
Voc fallback used	The digitised J–V does not reach $J = 0$. Extend the digitisation to higher voltages, or accept the fallback if the reported Voc is reliable.
$\lambda_{\text{onset},95\%}$ seems wrong	Check that the SPD CSV covers 400–750 nm or more. A truncated SPD will shift $\lambda_{\text{onset},95\%}$ inward.

12. Key Physical Constants Used

Constant	Value and Meaning
h (Planck constant)	$6.626 \times 10^{-34} \text{ J}\cdot\text{s}$
c (Speed of light)	$2.998 \times 10^8 \text{ m/s}$
q (Elementary charge)	$1.602 \times 10^{-19} \text{ C}$
K_m (Luminous efficacy)	683 lm/W — the maximum spectral luminous efficacy at 555 nm
$h\cdot c$ [J·nm]	$1.9864 \times 10^{-25} \text{ J}\cdot\text{nm}$ — used to convert photon flux to irradiance
TOL	10.0 % — the $\pm 10\%$ acceptance threshold for all three checks
TARGET_LUX	1000 lux — always fixed; do not change

13. Reference

The three-check validation protocol and the $\lambda_{\text{onset},95\%}$ spectral descriptor are based on:

Khampa et al. (2026). *"A locus-based benchmarking framework for indoor photovoltaics."* Newton (in press). DOI: [as cited in the project report].

Project report: *"A Universal Design and Benchmarking Framework for Indoor Photovoltaics."* Olarewaju Okikioluwa Victor & Ali Ghahremani. SEE 894, Simon Fraser University, Spring 2026.

This guide was generated from the source code of ipv_validation.m. All formulas, unit conversions, and parameter descriptions reflect the actual implementation in the script.